Rewrites Are Not Lessons

Teachable Edits as an Output Specification for Multimodal Language Learning

Wajdi Zaghouani Northwestern University in Qatar • wajdi.zaghouani@northwestern.edu

Multi-LLL 2026
Workshop on Multi-modality and Multi-lingual Language Learning
@ AACL-IJCNLP 2026

Position paper • no new experiments

A fluent rewrite is a correction, not a lesson. Tutors should output a teachable edit.

Localized on the learner's production, typed, conservative, explained or abstained, and scored as feedback.

Same page, two outputs

A CFL learner writes "I have a teacher"; the task asks for the polite form.

Learner page 我有一个老师
Wǒ yǒu yí gè lǎoshī

Correction engine 我有一位老师
Emits a fluent string and stops. Looks like a tutor in a screenshot.

Teachable edit 我有一个→位老师
Points at the classifier slot; 个 → 位; type: classifier / register. Leaves the rest untouched; explains the register contrast or abstains.

Arabic, same record. الطالبة مجتهد → مجتهدة: point at the adjective, type gender agreement, add the feminine ending. If the final handwritten character is ambiguous, coarsen the pointer or abstain.

Three positions

- **Capability is not instruction.** Reading a page and emitting a correct sentence does not show the model can teach.
- **Rewrites are not lessons.** Full-sentence generation hides the learner's interlanguage: what was already usable, and which form was the lesson.
- **Explanation is a first-class output.** A fluent wrong rule is worse than marking a span and stopping.

The teachable-edit record

One output record per edit. Each field is scored on its own.

Field	Chinese example	Evaluation
Pointer	box over the classifier slot on the page	box IoU sweep / span F_1; coarsening rate
Edit	个 → 位	edit P/R; pseudo-error rate
Error type	classifier / register	type accuracy (κ)
Target form	一位老师 *yí wèi lǎoshī*	match to references
Explanation	the polite classifier 位 matches the formal task requirement	see faithfulness
Faithfulness	covers this edit / related / unrelated	3-way rating; span grounding
Factuality	correct rule / invented / abstained	3-way rating; entailment
Level	elementary band	lexical band; teacher rating
Abstention	no rule if type unknown	coverage-risk; teacher-anchored
Conservatism	no other tokens changed	unnecessary-edit rate; budget

What is new: five jointly required properties

1. Localization into the original multimodal artifact
2. Conservatism as an evaluated property, not a prompting style
3. Explanation faithful to the proposed edit
4. Calibrated abstention
5. All fields in one record, so feedback quality is scored apart from correctness and uptake

Two-layer types. A language-specific surface layer (ERRANT, character-level CFL annotation) plus a small portable functional layer: form agreement, argument structure, selection and collocation, orthography and character formation, cohesion, pragmatic appropriateness.

Closest prior work covers parts

	EXPECT	EXCGEC	JELV	Ours
Edit-wise type	✔	✔	n/a	✔
Free-text explanation	limited	✔	–	✔
Edit-validity test	–	–	✔	✔
Pointer to artifact	–	–	–	✔
Scored abstention	–	–	–	✔
Split evaluation targets	–	–	–	✔

Prior instruments are text-only or validity-only. The specification is the joint record.

Four targets, reported separately

Correction correctness
Edit matches a reference repair. $F_{0.5}$ fits here.

Feedback quality
Pointer, type, conservatism, explanation, abstention

Immediate uptake
Learner uses it on a later attempt, same session

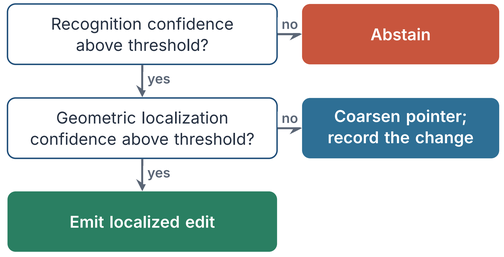
Learning outcomes
Form retained or transferred to new contexts

What M2/ERRANT $F_{0.5}$ cannot see: pedagogical appropriateness, explanation faithfulness, proficiency alignment, localization, uptake.
A correct string with a wrong box is a failed lesson and a successful $F_{0.5}$ item.

VisCGEC baseline (OCR + fine-tuned Qwen): 28.59 character-level and 15.85 sentence-level $F_{0.5}$. A capability result on a hard visual task, not a learning result.

Localization under recognition noise

The pointer is the load-bearing multimodal field. A two-stage gate, in order:



Recognition failures are logged as a separate **pseudo-error rate**, never shown as learner errors. For Arabic, keep text-span order and image-coordinate order distinct, and state script direction.

Retrofit, don't rebuild

VisCGEC (2,451 handwritten CFL images) already annotates boxes, corrections, and error types: **three fields are scorable today**. Still missing: explanation, level, conservatism, abstention.

Checklist for workshop chairs

1. Add feedback-quality and, where possible, uptake columns, without changing $F_{0.5}$ rankings
2. Ask system papers that claim educational use to report the record
3. Reserve a session for systems with high $F_{0.5}$ but low explanation faithfulness

Same record on speech. The pointer becomes a timestamp; ASR-induced edits count as pseudo-errors.

Stated failure condition: in a small double-annotated pilot, Krippendorff's $\alpha < 0.67$ on faithfulness means the rubric needs revision.

Limitations: position paper; no evidence yet that teachable edits improve uptake over rewrites; transfer to speech, slides and other scripts not shown empirically; the Chinese and Arabic examples are constructed, not benchmark items

Questions? wajdi.zaghouani@northwestern.edu